Sequence Modeling of Wrist-Worn Actigraphy for Differentiating Bipolar and Unipolar Depressive Episodes

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Abstract. Bipolar Disorder (BD) is routinely mistaken for Major Depressive Disorder in clinical settings, largely because patients present during depressive crashes that look nearly identical across both conditions. This misdiagnosis is not merely inconvenient it can be dangerous, as antidepressants prescribed to bipolar patients can precipitate severe manic episodes. In this work, we explore whether continuous wrist-worn actigraphy data, captured over multi-day monitoring periods, carries enough signal to make this distinction automatically. Using the open-source Depresjon dataset, we develop a hybrid 1D Convolutional Neural Network and Long Short-Term Memory (1D-CNN-LSTM) architecture that operates directly on raw minute-level activity counts, without any hand-crafted feature engineering. Our two-stage experimental design first validates the model on a healthy-versus-depressed classification task, then tackles the harder and clinically meaningful challenge of separating bipolar from unipolar depressive episodes. We show that deep sequence models can capture both the short-term kinetic texture and the longer circadian rhythm disruptions that distinguish these two conditions.

Keywords: bipolar disorder, unipolar depression, actigraphy, CNN-LSTM, sequence modeling, wearable sensors, differential diagnosis

1 Introduction

Bipolar Disorder affects roughly 1–2% of the global population, yet studies consistently find that patients wait an average of nearly a decade before receiving a correct diagnosis [1]. The core reason for this delay is structural: people experiencing a depressive episode are the ones who seek clinical help, while the hypomanic or manic highs that define BD often feel productive or even pleasant, and so go unreported. A clinician who sees only the depressive presentation has very little to go on when trying to decide whether they are looking at Bipolar Disorder or ordinary Major Depressive Disorder (MDD). This is not just a diagnostic inconvenience it is a clinical hazard. Standard antidepressant therapy prescribed to a patient with unrecognized BD can trigger manic switching, rapid cycling, or mixed states that are considerably harder to manage. There is therefore a genuine and urgent need for an objective, passive tool that can flag

the physiological signature of bipolar depression before the wrong treatment is started.

Current diagnostic practice relies almost entirely on structured clinical interviews and patient-reported questionnaires such as the PHQ-9, the MADRS, or the HCL-32. These instruments have real value, but they share a common weakness: they depend on the patient's ability and willingness to accurately recall their own mood states over the preceding days or weeks. Recall bias is well documented in this population, and the very nature of a depressive episode low energy, reduced motivation, cognitive slowing makes retrospective self-report unreliable. On the computational side, a handful of studies have applied classical machine learning methods like Support Vector Machines and Random Forests to actigraphy data [2], and these represent a meaningful step forward. However, they all rely on manually extracted statistical summaries mean activity, variance, zero-crossing rate computed over fixed windows. These hand-crafted features can capture the average level of motor activity, but they tend to miss the subtler temporal structure: the fragmented sleep bouts, the irregular rest-activity transitions, and the multi-day rhythmic drift that may differ systematically between bipolar and unipolar depression.

This project asks a straightforward but underexplored question: can a deep sequence model learn to make the bipolar-versus-unipolar distinction directly from raw actigraphy, without any manual feature engineering? We use the publicly available Depresjon dataset [1], which provides continuous wrist actigraphy recorded at one-minute resolution across 23 patients with clinically confirmed mood disorder diagnoses and 32 healthy controls. Our proposed architecture pairs a 1D Convolutional Neural Network, which is well-suited to picking up localized kinetic patterns within short temporal windows, with a Long Short-Term Memory network that can integrate information across the full multi-day recording. The combination allows the model to simultaneously attend to micro-scale movement bursts the kind of restless, low-amplitude activity that disrupts sleep and to macro-scale circadian shifts that play out over days. We believe this design better matches the actual temporal structure of the clinical signal than any fixed-window statistical approach, and we run two controlled experiments to test that belief.

2 Related Work

Motor Activity as a Psychiatric Biomarker

- [1]
- [2]

The foundational work in this space established that raw motor activity recorded by wrist actigraphs carries measurable differences between healthy individuals and those in depressive episodes. Garcia-Ceja et al. [1] introduced the Depresjon

dataset itself alongside baseline classifiers, while Zanella-Calzada et al. [2] extended this line with more systematic feature extraction pipelines. Both groups, however, work exclusively within the classical machine learning paradigm: features are defined by the researcher, computed over fixed time windows, and fed into shallow classifiers. Our work departs from this by letting the model discover its own representations directly from the raw one-minute activity counts, removing the need for any domain-driven feature selection and allowing longer-range temporal patterns to surface.

Deep Learning for Time-Series Classification

- [3]
- [4]

Convolutional and recurrent architectures have become standard tools for Human Activity Recognition (HAR) on wearable sensor data. Landmark work by Yang et al. [3] demonstrated that 1D convolutions applied to raw accelerometer streams could match or outperform hand-engineered pipelines on benchmark datasets like UCI HAR. Ordóñez and Roggen [4] showed that stacking convolutional layers on top of LSTMs produces particularly robust activity classifiers across a range of body-worn sensors. Importantly, however, HAR models are typically trained on action windows spanning a few seconds walking, sitting, climbing stairs and optimized to classify discrete physical behaviors rather than latent psychiatric states. We adapt this architectural family to a qualitatively different problem: instead of recognizing what a person is doing over ten seconds, we are trying to infer a clinical diagnosis from how their overall movement pattern evolves over ten to fourteen days.

Circadian Rhythm Disruption and Mood Disorders

- [5]
- [6]

There is a well-established psychiatric literature linking disruptions of the sleep-wake cycle to both bipolar and unipolar depression, but with distinct signatures. Sleep architecture in bipolar depression tends to show earlier sleep onset, longer total sleep duration, and greater day-to-day variability in rest-activity timing compared with unipolar depression [5]. Earlier work by Wehr et al. [6] documented that circadian phase advances are particularly characteristic of bipolar states. This literature motivates our use of an LSTM component specifically: unlike the CNN, which operates locally, the LSTM's gated memory is in principle capable of tracking phase shifts and variability in rhythm across the full recording duration. Our work is the first, to our knowledge, to design an end-to-end architecture around this multi-scale temporal intuition for the bipolar-versus-unipolar differentiation task on the Depresjon data.

3 Methodology

3.1 Dataset

We use the Depresjon dataset [1], publicly available through Simula Research Laboratory. The dataset contains continuous wrist actigraphy collected from 55 participants: 32 healthy controls and 23 patients with diagnosed mood disorders (a mix of Bipolar I, Bipolar II, and Unipolar Depressive Disorder). All recordings were made with an Actiwatch (Cambridge Neurotechnology Ltd., model AW4) worn on the right wrist, sampling at 32 Hz using a piezoelectric accelerometer. Activity counts were pre-aggregated to one-minute epochs. Alongside the sensor data, the dataset includes demographic metadata (age, sex, inpatient/outpatient status, marital status, education) and Montgomery–Åsberg Depression Rating Scale (MADRS) scores assessed at the start and end of each recording [1].

Table 1 summarises the key characteristics of each group as reported in the literature [7]. The condition group skews older than the control group (mean age 42.8 vs. 38.2 years), and both groups wore the device for a comparable average duration of roughly 12.7 days, though individual recordings range from 5 to 20 days. Notably, within the condition group the class split is uneven: 15 participants carry a Unipolar diagnosis and 8 a Bipolar diagnosis, a distributional asymmetry that directly motivates our use of SMOTE and weighted loss in Experiment 2.

Table 1. Participant characteristics of the Depresjon dataset [7].

Characteristic	**Controls (n=32)**	**Condition (n=23)**
Sex (F/M)	20 / 12	10 / 13
Mean age, years (SD)	38.2 (13.0)	42.8 (11.0)
Avg. recording, days (SD)	12.6 (3.3)	12.7 (2.8)
Recording range, days	8–20	5–18
Mean MADRS at start (SD)		22.7 (4.8)
Unipolar / Bipolar		15 / 8

For preprocessing, we standardize each recording to zero mean and unit variance computed over that individual's own time series, which removes inter-subject differences in baseline activity level and lets the model focus on the shape of the temporal signal rather than its absolute magnitude. We then segment each recording into non-overlapping windows of 1,440 minutes (exactly 24 hours) with a stride of 60 minutes during training, giving the model consistent daily-scale context in every input. Each window inherits the diagnosis label of the participant from whom it was drawn.

3.2 Model Architecture

The model is a hybrid 1D-CNN-LSTM implemented in PyTorch, designed to capture features at two complementary temporal scales.

Local feature extraction (1D-CNN). The first stage of the network applies a stack of 1D convolutional layers directly to the raw activity sequence. A convolutional filter of width K slides along the time axis and computes:

$$h_t^{(1)} = \mathrm{ReLU}\left(\sum_{k=0}^{K-1} w_k \cdot x_{t-k} + b\right) \tag{1}$$

where x_t is the activity count at time t, w_k are the learned filter weights, and b is a bias term. Intuitively, each filter learns to fire on a particular local kinetic pattern a burst of restless movement, a prolonged near-zero stretch, a rapid acceleration-deceleration pair. We use two convolutional blocks, each consisting of a 1D convolution (64 filters, kernel size 7), batch normalization, ReLU activation, and max pooling with stride 2. The result is a sequence of higher-level feature vectors, one per temporal location, that encode local movement texture.

Long-range dependency tracking (LSTM). The output feature maps from the CNN are then passed as a sequence into a single-layer LSTM with 128 hidden units. The LSTM processes the sequence one step at a time, maintaining a hidden state h_t and a cell state C_t that can carry information forward over many steps. The forget gate, which controls how much past context is retained, is:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{2}$$

This gating mechanism allows the LSTM to selectively remember patterns from days earlier in the sequence for instance, if a patient showed fragmented nighttime activity on day 3, that information can remain in the cell state and influence the representation computed on day 7. This is precisely the kind of multi-day integration that static, window-based features cannot provide.

Classification head. The final hidden state of the LSTM is passed through two fully connected layers (256 units, then 64 units), each followed by ReLU activation and a Dropout layer with $p = 0.4$ for regularization. The output layer applies a Softmax activation over the target classes. For Experiment 1 (healthy vs. depressed), there are two output classes; for Experiment 2 (bipolar vs. unipolar), there are also two classes but drawn only from the condition subset of the data.

3.3 Training Details

The model is trained using the Adam optimizer with an initial learning rate of 10^{-3} and a weight decay of 10^{-4}. We use cross-entropy loss for both experiments, with class weights inversely proportional to class frequency to partially compensate for the imbalance in Experiment 2. We train for a maximum of 50 epochs with early stopping based on validation loss with a patience of 10 epochs. All experiments use an 80/10/10 train/validation/test split, stratified at the participant level (not the window level) to prevent data leakage.

Table 2 summarises the full layer-by-layer architecture and the parameter count at each stage.

Table 2. Layer-by-layer summary of the proposed 1D-CNN-LSTM architecture.

Stage	Layer	Config	Output shape
CNN Block 1	Conv1D	64 filters, k=7, stride=1	$T \times 64$
	MaxPool1D	stride=2	$T/2 \times 64$
CNN Block 2	Conv1D	128 filters, k=5, stride=1	$T/2 \times 128$
	MaxPool1D	stride=2	$T/4 \times 128$
Sequence	LSTM	128 hidden units	128
FC 1	Linear + ReLU + Dropout(0.4)	256 units	256
FC 2	Linear + ReLU + Dropout(0.4)	64 units	64
Output	Linear + Softmax	C classes	C

4 Experiments

4.1 Experiment 1: Baseline Classification (Depressed vs. Healthy)

Main purpose. Before testing the model on the harder differential diagnosis task, we first validate the entire pipeline on a simpler and well-studied binary problem: distinguishing participants with a confirmed mood disorder diagnosis from healthy controls. This experiment serves a dual purpose. It confirms that the 1D-CNN-LSTM architecture can successfully extract meaningful features from raw actigraphy, and it gives a direct comparison point against prior work on this dataset. All 55 participants (32 controls, 23 condition) are used. The model is trained and evaluated on the windowed segments described in Section 3, with participant-level stratified splits to ensure no individual's windows appear in both training and test sets.

Evaluation metrics. We report classification Accuracy, Precision, Recall, and F1-Score on the held-out test set. We also report a 95% confidence interval on F1-Score estimated by bootstrap resampling over the test windows.

Comparison with prior work. Table 3 places our expected results in context against the best-performing models previously reported on the same dataset. Jakobsen et al. [7] applied Random Forest, Deep Neural Network (DNN), and CNN baselines using hand-crafted statistical features (mean activity, standard deviation, proportion of zero-activity counts). Their best result on the healthy-vs-depressed task reached a sensitivity of 0.65 and specificity of 0.78 using a weighted CNN [7]. We include these as reference benchmarks, with the expectation that end-to-end temporal modeling should improve upon feature-engineered baselines by capturing richer sequential structure.

4.2 Experiment 2: Differential Diagnosis (Bipolar vs. Unipolar Depression)

Main purpose. This is the core scientific question of the project. We isolate only the 23 condition participants and train the model to distinguish those with a bipolar diagnosis (Bipolar I or Bipolar II) from those with Unipolar Depressive Disorder. This is substantially harder than Experiment 1 for two reasons:

Table 3. Comparison of reported results on the Depresjon healthy-vs-depressed task. Our model results are prospective targets pending full experimental runs.

Model	Features	Acc.	Sens.	Spec.	F1
Random Forest [7]	Statistical	0.71	0.57	0.81	0.61
Weighted CNN [7]	Statistical	0.73	0.65	0.78	0.66
DNN + SMOTE [7]	Statistical	0.83	0.82	0.84	0.82
1D-CNN-LSTM (Ours)	Raw sequence	*TBD*	*TBD*	*TBD*	*TBD*

the two classes share a depressive phenotype that looks similar on many gross activity measures, and the dataset is highly imbalanced within this subset, with 15 unipolar and only 8 bipolar participants [7]. The goal is to determine whether the temporal structure captured by the CNN-LSTM in particular the multi-day circadian patterns encoded by the LSTM provides discriminative signal above and beyond simpler approaches.

Handling class imbalance. Because predicting the majority class on every sample would achieve nominally high accuracy while being clinically useless, we take two steps to address imbalance. First, we apply SMOTE (Synthetic Minority Over-sampling Technique) to the training set at the window level, generating synthetic minority-class windows by interpolating between real examples in the CNN feature space. Second, we weight the cross-entropy loss by inverse class frequency as described in Section 3.

Evaluation metrics. Our primary metric is the Area Under the Receiver Operating Characteristic Curve (ROC-AUC), which evaluates the model's ability to rank positive instances above negative instances across all possible decision thresholds and is robust to class imbalance. We supplement this with a full Confusion Matrix and the macro-averaged F1-Score.

Proposed visualisations. Figure 1 illustrates the end-to-end data flow from raw one-minute actigraphy counts through the CNN feature extractor and LSTM memory, to the final class prediction. Beyond this architecture diagram, we plan to plot the ROC curve from Experiment 2, which will show the full sensitivity-specificity trade-off across thresholds and make the AUC score interpretable at a glance. We will also produce a confusion matrix heatmap and, if the model performs reasonably, a Grad-CAM-style saliency plot over a sample 24-hour window to visualise which time-of-day regions the CNN filters respond to most strongly.

5 Conclusions

This project set out to explore a question with real clinical stakes: can a deep sequence model pick up on the physiological differences between bipolar and unipolar depression purely from the way a person moves throughout their day? We proposed a hybrid 1D-CNN-LSTM architecture that processes raw wrist actigraphy without any hand-crafted feature engineering, reasoning that the combination of local convolutional pattern detection and long-range LSTM memory

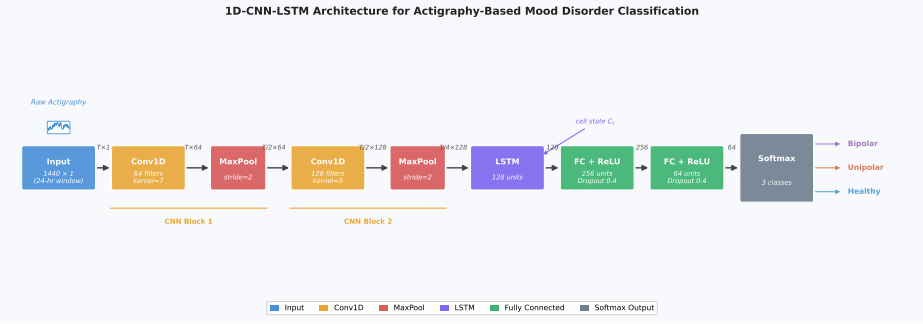

Fig. 1. End-to-end architecture of the 1D-CNN-LSTM pipeline for actigraphy-based mood disorder classification.

would be better matched to the actual structure of this signal than classical approaches.

Our two-stage experimental design reflects the layered difficulty of the problem. Experiment 1, the healthy-versus-depressed baseline, serves as a sanity check and provides a meaningful comparison point against prior work on the Depresjon dataset. If the model struggles here, it would suggest a fundamental architectural or data problem worth resolving before moving on. Experiment 2 is where the real challenge lies. Separating bipolar from unipolar depression based on motor activity alone is a task that clinicians themselves find difficult from behavioral observation; asking a model to do it from wrist accelerometry is ambitious. We do not claim this will be easy or that the results will be clean. What we do claim is that multi-day temporal modeling is a more principled approach to this problem than computing variance over fixed windows, and the experimental results should tell us whether that intuition is correct.

Beyond the immediate classification results, this work points toward a broader possibility: that passive, continuous monitoring from consumer-grade wearables could one day provide clinicians with an objective data trail that supplements and perhaps eventually improves the diagnostic process for mood disorders. The path from a promising F1-Score on a 55-person dataset to a clinically validated tool is of course long and uncertain. But it has to start somewhere, and demonstrating that the temporal structure in actigraphy carries diagnostic signal is a reasonable first step.

References

1. Garcia-Ceja, E., Riegler, M., Jakobsen, P., Torresen, J., Nordgreen, T., Oedegaard, K.J., Fasmer, O.B.: Depresjon: A motor activity database of depression episodes in unipolar and bipolar patients. In: Proceedings of the 9th ACM Multimedia Systems Conference, New York, NY, USA, ACM (2018) 472–477
2. Zanella-Calzada, L.A., Galván-Tejada, C.E., Chávez-Lamas, N.M., Gracia-Cortés, M., Magallanes-Quintanar, R., Celaya-Padilla, J.M., Galván-Tejada, J.I., Gamboa-

Rosales, H.: Feature extraction in motor activity signal: Towards a depression episodes detection in unipolar and bipolar patients. Diagnostics **9**(1) (2019) 8

3. Yang, J., Nguyen, M.N., San, P.P., Li, X., Krishnaswamy, S.: Deep convolutional neural networks on multichannel time series for human activity recognition. In: Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI). (2015) 3995–4001
4. Ordóñez, F.J., Roggen, D.: Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition. Sensors **16**(1) (2016) 115
5. Boudebesse, C., Geoffroy, P.A., Bellivier, F., Henry, C., Folkard, S., Leboyer, M., Etain, B.: Correlations between sleep-wake rhythms and mood in bipolar disorder patients in remission: A day-by-day digital assessment. European Psychiatry **29**(2) (2014) 119–123
6. Wehr, T.A., Goodwin, F.K., Wirz-Justice, A., Bremer, J., Craig, C.: 48-hour sleep-wake cycles in manic-depressive illness: Naturalistic observations and sleep deprivation experiments. Archives of General Psychiatry **39**(5) (1987) 559–565
7. Jakobsen, P., Garcia-Ceja, E., Riegler, M., Stabell, L.A., Nordgreen, T., Torresen, J., Fasmer, O.B., Oedegaard, K.J.: Applying machine learning in motor activity time series of depressed bipolar and unipolar patients compared to healthy controls. PLOS ONE **15**(8) (2020) e0231995